MILITARY SPECIFICATION

TELESCOPE, ELBOW: M118 SERIES AND T176E4

This amendment forms a part of Military Specification MIL-T-46316B(MU) dated 10 July 1968.

Para 1.1: Delete and substitute: "This specification covers similar types of instruments designated as Telescope, Elbow: M118, M118C, M118CA1 and T176E4. These telescopes are basically identical except for differences in reticle patterns."

Para 2.1: Under Drawings, insert: "M118CA1" after M118C. Under Packaging Data Sheet first line, delete: "og" and substitute "of". Under Packaging Data Sheet, add: "10559855" and "Packaging of Telescope, Elbow: M118CA1".

Para 3.3.1. Delete and substitute: "Storage temperatures.- The telescope shall show no evidence of physical failure at room temperature (60° to 90°F) after having been thermally stabilized at -70°F and +160°F."

Para 3.3.2. Delete and substitute: "Standard ambient temperatures.- Subsequent to meeting the requirements of 3.3.1, the telescope shall meet all the requirements of this specification except 3.3.3, at a standard ambient temperature between 60°F and 90°F."

Para 3.3.3. Delete and substitute: "Operating Temperatures.- The telescope shall show no evidence of physical failure and shall meet the applicable requirements of 3.9 while thermally stabilized at -40°F and +150°F."

Para 3.3.4. Insert new paragraph: "3.3.4 Vibration.- The telescope shall show no evidence of physical failure (loose, missing or broken parts) after having been vibrated at a frequency of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds. Vibration shall not have caused the eyepiece arm to disengage from the initially engaged position. Subsequent to vibration, the telescope shall meet the requirements of 3.4 to 3.10 inclusive.

Para 3.5. Delete and substitute: "Sealing.- The telescope shall maintain an internal pressure of dry nitrogen at 5 pounds per square inch (psig) for a minimum period of 2 hours without loss of pressure in excess of 0.2 psig. The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F."

Para 3.6. Second line, delete: "kingpin" and substitute: "tapered."

Para 3.7.1. Second line, delete: "the reticle" and substitute: "both reticles."

Para 3.7.5. Delete and substitute: "Boresight error due to reticle rotation.- The line of sight at the boresight point shall not deviate more

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than 0.25 mil when the reticle is canted a minimum of 5 degrees to the left and a minimum of 5 degrees to the right by means of the cant correction knob."

Para 3.9. Fifth line, delete: "3.3.2" and substitute "3.3.3"

Para 3.9.1. Second line delete: "kingpin" and substitute: "tapered."

Para 3.9.3. Delete and substitute: "Mirror, plunger, arm and rheostat.- The level vial mirror (pivotal type only), plunger knob on the reticle shift lever, eyepiece arm and the rheostat knob shall be mechanically operable when exposed to the operating temperatures of 3.3.3."

Para 4.3.2.1. Delete from first sentence: "Listed in the list of inspection documents (IEL)."

Table I. Change: "3.3.3" to "3.3.4."

Table II. Change: "3.3.2" to "3.3.3."

Para 4.6.1.1. Change: "3.3.1 and 3.3.2" to "3.3.1 3.3.2 and 3.3.3" in two places.

Para 4.7.2.2. Change: "3.3.3" to "3.3.4" in two places.

Para 5.1. Add following: "10559855 Telescope Elbow M118CA1"

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